

Which Damage Do the Images Support? Evidence-Bounded 3D Damage Estimation on Turbine Blades

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Abstract

001 Visual inspection of aero-engine blades must decide
002 whether a change seen in an image is physical damage,
003 where it is on the part, and how deep it goes. Ex-
004 isting methods flag anomalies in 2D, output boxes or
005 voxels, or require a measured 3D scan, and none states
006 whether the available views are sufficient to support
007 a claimed shape change. We study evidence-bounded
008 3D damage estimation: given a healthy CAD model
009 and a few calibrated images, output a displacement
010 field on the blade surface when the images support one,
011 and abstain when camera error, lighting or a surface-
012 colour change explain the images equally well. Our
013 estimator compares a nuisance-only explanation (per-
014 view pose, light gains, an albedo patch) with the same
015 explanation plus a continuous dent, using a differen-
016 tiable renderer with a neural shader. Dent hypotheses
017 are proposed by a Rao score test computed for every
018 surface location in one backward pass, and the deci-
019 sion threshold on the resulting fit gain is calibrated on
020 healthy and colour-only blades with a split-conformal
021 guarantee. We release BladeDelta, a benchmark of 696
022 rendered inspection cases on 84 public Rotor37 blade
023 geometries, with exact damaged meshes, colour-only
024 negatives, three dent families, controlled capture per-
025 turbations and two sealed test splits, plus 456 cases on
026 six industrial blade designs. On a sealed test of 30 un-
027 seen blades the method locates 54% of visible dents (12
028 of 13 deeper than 0.2 mm; median depth error 0.03 mm)
029 at a 10% false-alarm rate, while CAD-prior deformation
030 fields and a lifted 2D detector locate at most 16%;
031 a second sealed test replicates the result (54% found,
032 6% false alarms). On the unseen industrial designs,
033 correcting the healthy reference for silhouette and cam-
034 era motion raises AUROC from 0.72 to 0.95 and cuts
035 false alarms from 51% to 4%.

1. Introduction

Turbine and compressor blades are inspected with cam-
eras, and the inspector’s question is physical: is there
a dent, where on the blade is it, and how deep is it?
Damage of interest is small. On a blade imaged at
 512×512 pixels, a dent of 0.07 mm changes a few hun-
dred pixels by two to seven grey levels. A one-degree
camera error, a ten-percent lighting change or a stain
changes thousands.

Current vision methods answer an easier question.
Industrial anomaly detection localises unusual pix-
els [1, 18], including under pose and illumination
change [3, 11]. Reference-based methods compare a
photograph with a CAD model and output boxes [2], or
lift per-view scores into an anomaly volume [7]. CAD-
conditioned reconstruction fits a deformation field to
sparse posed views [17, 21], and leaves measurement-
quality damaged surfaces to future work [17]. None
of these outputs states whether the images actually
support a physical change, as opposed to a nuisance
explanation.

Our own earlier experiments made this concrete. A
render-and-compare fit on public Rotor37 blade geom-
etry produced false dents on 14 of 14 healthy or colour-
only blades under a one-degree camera error, and on
unfamiliar dents the best image match selected a worse
3D shape in 8 of 12 depth sweeps. Better image agree-
ment is not better geometry, and a plausible recon-
struction can choose the wrong physical explanation.

We therefore formulate evidence-bounded 3D dam-
age estimation. The input is a healthy CAD mesh and
 K calibrated images; the output is a per-vertex dis-
placement field together with a decision, supported or
not supported by these views. Our estimator treats
this as a nested hypothesis comparison. H_0 explains
the images with nuisance only: per-view pose correc-
tions, light gains and a surface albedo patch. H_1 adds a
continuous dent. The fit gain $L(H_0) - L(H_1)$ is thresh-
olded at a value calibrated on negatives, giving a con-

075	trolled false-change rate.	
076	Making this work required separating the roles of	
077	the evidence. Pose is fitted from silhouettes, which do	
078	not depend on lighting; the dent is compared against	
079	nuisance only on interior surface pixels that are not	
080	grazing, not near a silhouette and not on the blade	
081	root. Dent hypotheses come from a Rao score test	
082	evaluated for every surface location and radius with	
083	one backward pass through the renderer, and the dent	
084	and colour-patch explanations are searched symmetrically,	
085	so a colour patch fitted first cannot hide a dent.	
086	We also report observability: on our development split,	
087	70% of dents placed on camera-facing surfaces are less	
088	than half visible in the two input views, and the correct	
089	output for those is to abstain.	
090	Contributions:	
091	• The task of evidence-bounded 3D damage estimation	
092	from sparse views and healthy CAD, with metrics	
093	that separate localisation, shape accuracy, false	
094	physical change and abstention on unobservable	
095	damage.	
096	• An estimator combining a differentiable renderer	
097	with a neural shader, silhouette-only pose, interior-	
098	domain nested hypothesis testing, score-test proposals	
099	and split-conformal calibration.	
100	• BladeDelta, a benchmark built on public Rotor37	
101	geometry [4] with exact damaged meshes, colour-	
102	only negatives, three dent families that differ from	
103	the estimator's own profile, and controlled camera	
104	and lighting perturbations, with a protocol and	
105	sealed test split fixed before evaluation.	
106	2. Related Work	
107	Industrial and 3D anomaly detection. Memory-bank	
108	methods such as PatchCore [18] localise anomalous	
109	pixels; MVTec 3D-AD [1] and Anomaly-ShapeNet [13]	
110	add measured or synthetic 3D. Pose-agnostic settings	
111	use multi-pose benchmarks [23] and Gaussian-splat	
112	references [11], and M ² AD [3] benchmarks view-	
113	illumination interplay. Multi-view settings include	
114	RAD [5] and SiM3D [7], which lifts per-view scores	
115	into an anomaly volume; recent work warns that adding	
116	views can leak information in multi-view evaluation [6].	
117	Synthetic anomalies have been argued sufficient for 3D	
118	detection [20], and incomplete-modality methods train	
119	with 3D and test on RGB [19]. These methods output	
120	anomaly scores; they do not estimate a physical surface	
121	change or state whether the views support one.	
122	Reference CAD and blade inspection. Looking 3D [2]	
123	aligns a query photograph with a reference shape and	
124	localises abnormal regions with boxes. BladeSynth [8]	
125	renders procedural nicks, dents, scratches and corrosion	
	on aero-engine blades for 2D detection. Model-free uni-	126
	versal detection reconstructs a normal shape by GAN	127
	inversion [15]. We instead output a 3D displacement	128
	field on a known design.	129
	CAD-conditioned reconstruction. CAD-NeRF [21] re-	130
	trieves a CAD model and jointly optimises deformation,	131
	pose and appearance from few uncalibrated views.	132
	CADSplat [17] fits a shared deformation field to sparse	133
	posed views with a CAD prior and explicitly leaves	134
	measurement-quality deformed surfaces to future work.	135
	Our free-form deformation baseline follows this family.	136
	The difference is the output: a decision on whether	137
	the deformation is supported, with a calibrated false-	138
	change rate.	139
	Inverse rendering and uncertainty. Inverse rendering	140
	separates shape and material from images [16, 22];	141
	we use differentiable rasterisation [12] with a learned	142
	shader for the same purpose on a known geometry.	143
	Radiance-field uncertainty [9, 10] and observation com-	144
	pleteness [14] quantify where a reconstruction is unre-	145
	liable. They do not test a physical-change hypothesis	146
	against nuisance explanations, which is the question an	147
	inspector needs answered.	148
	3. Method	149
	Setup. We are given a healthy mesh $\mathcal{M} = (V, F)$ with	150
	vertices $V \in \mathbb{R}^{N \times 3}$, K observed images $I_1, \dots, I_K$ with	151
	nominal intrinsics and extrinsics, and nominal light-	152
	ing. We seek a displacement field $D \in \mathbb{R}^{N \times 3}$ on the	153
	airfoil and a decision $s \in \{\text{supported}, \text{not supported}\}$.	154
	Nuisance variables η are a per-view pose correction (a	155
	rotation about the part and a camera-frame translation),	156
	three light gains, and an optional albedo patch.	157
	Differentiable renderer. We rasterise $\mathcal{M} + D$ with nvd-	158
	iffrastr [12] using smooth vertex normals. A neural	159
	shader maps the per-pixel world normal n , position x	160
	and view direction ω to one radiance component per	161
	light, $R_k(n, x, \omega)$, and an environment term R_0 . Be-	162
	cause light transport is linear in each light's intensity,	163
	the pixel colour is $T(a \odot (R_0 + \sum_k g_k R_k))$ for albedo	164
	multiplier a , light gains g and a learned monotone tone	165
	curve T , followed by anti-aliasing and the sensor's pixel	166
	filter. An unstructured shader that took g as an input	167
	learned to ignore it, and lighting changes then pro-	168
	duced spurious dents. The shader is trained once, on	169
	renders of blade geometries disjoint from all evaluation	170
	geometries, with a paired loss on image differences be-	171
	tween lighting settings. The healthy CAD rendered under	172
	the nominal capture provides a per-pixel correction	173

174 $B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$ that absorbs static shader error.
 175 On a held-out blade its mean error is 0.92 grey levels;
 176 image changes caused by a lighting change correlate
 177 with the reference renderer at 0.90–0.98, and changes
 178 caused by a dent at 0.96–0.98 in the 120° view.

179 Nuisance fit (H_0). Each view’s pose is initialised by
 180 Euclidean ECC alignment of the observed and refer-
 181 ence images, then refined by matching anti-aliased sil-
 182 houettes to the observed foreground, coarse to fine. Sil-
 183 houettes do not depend on lighting, which prevents the
 184 pose from compensating for a brightness change. With
 185 pose fixed, light gains are fitted on the full image.

186 Evidence domain. All further comparisons use a fixed
 187 interior domain Ω : airfoil pixels at least 6px from any
 188 silhouette or self-occlusion edge, excluding the blade
 189 root and pixels where $|n \cdot \omega| < 0.35$. Both hypothe-
 190 ses are evaluated on the same Ω , so their losses are
 191 comparable.

192 Dent model (H_1). A dent is parameterised by a sub-
 193 vertex centre c (an anchor vertex plus a tangent offset),
 194 depth $\delta \geq 0$, radius r , aspect ratio, orientation, profile
 195 exponent p and a signed rim coefficient κ :

$$196 \quad D_i = -\delta \left[(1 - s_i^2)_+^p - \frac{\kappa}{2} \rho(s_i) \right] n_c, \quad (1)$$

197 where s_i is the anisotropic normalised distance of ver-
 198 tex i from c in the tangent plane, ρ a ring profile and n_c
 199 the centre normal, restricted to same-side airfoil ver-
 200 tices. None of the benchmark’s truth families matches
 201 this profile exactly.

202 Score-test proposals. Let $h \in \mathbb{R}^N$ be an inward nor-
 203 mal displacement. One backward pass gives $g =$
 204 $-\partial L_\Omega / \partial h$ at $h = 0$. For a kernel $w_{c,r}$ centred at c
 205 with radius r , the normalised score $\langle w_{c,r}, g \rangle / \|w_{c,r}\|$ is
 206 the Rao score statistic for a small dent at (c, r) against
 207 H_0 . We evaluate it for all candidate centres and two
 208 radii with a sparse matrix product, and start H_1 fits
 209 from the top- k after non-maximum suppression. Pro-
 210 posals are computed from the nuisance-only residual,
 211 and each start is fitted both with and without the H_0
 212 albedo patch, so a colour explanation fitted first cannot
 213 hide a dent.

214 Decision. The fit gain is $\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0$,
 215 as H_1 nests H_0 . We report D if $\Delta > \tau$ and $D = 0$ other-
 216 wise, with τ the split-conformal $(1-\alpha)$ quantile of Δ on
 217 calibration negatives (healthy and colour-only blades),
 218 which bounds the false-change rate on exchangeable
 219 negatives by α . We use a Charbonnier loss ($\epsilon = 2/255$)

Split	Blades	Healthy	Stain	Dent	Dent + stain
Shader training	13	only for the renderer; never evaluated			
Development	12	12	24	36	24
Calibration	12	24	36	36	24
Test (sealed)	30	30	60	90	60

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

summed over Ω ; a Student- t noise model and a wider Charbonnier floor did not improve separation on development data.

4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [4] (CC BY-SA 4.0), normalised to unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split; 67–109 are held in reserve.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in $[0.0005, 0.006]$ blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in $[0.04, 0.12]$. Centres face at least one of seven cameras and lie at least $\max(0.1, 1.2r)$ from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping, 512×512) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in $\pm 1.5^\circ$ per view); light jitter (each of three area lights $\times [0.85, 1.15]$); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered; protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ($\pm 0.5^\circ$, $\pm 1^\circ$, $\pm 1.5^\circ$). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in $[0.1, 1.0]$ mm and radius in $[1, 5]$ mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than 2×10^{-5}) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

5. Experiments

Baselines. Unchanged CAD reports the healthy mesh. Discrete search re-implements our earlier method: image-guided proposals, a fixed cubic profile of radius 0.08 and depths $\{0.0005, 0.001, 0.0015\}$, at nominal pose. Free-form field fits a per-vertex normal displacement with Laplacian smoothness and ℓ_1 sparsity after the same pose and lighting fit. MLP deformation field replaces the per-vertex values with a coordinate MLP (positional encoding, two hidden layers), the continuous deformation field used by CAD-prior reconstruction. These two baselines reimplement the core idea of CADsplat [17] — a smooth deformation field fit to sparse posed views under a CAD prior — within our renderer; they are not the official model, whose code had not been released at the time of writing. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate exposed blades to segment dents; detections are back-projected onto the mesh through the nominal cameras. All methods share the renderer, cameras and observations, and each is thresholded by the same calibration procedure on its own score.

Main results.

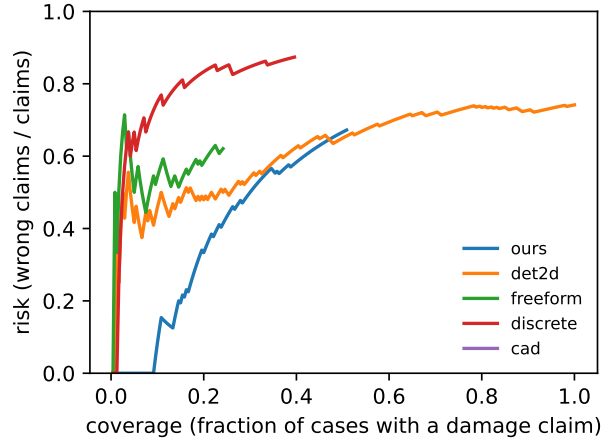


Figure 1. Risk (share of damage claims that are wrong) against coverage (share of cases with a claim) as the threshold varies, sealed test.

Risk-coverage.

Ablations.

Why recall is limited. Recall is set by dent depth and by the false-alarm budget, not by a failure to locate (Table 5). Every reported dent is in the right place, and 12/13 visible dents deeper than 0.2 mm are reported; most misses are shallower than 0.1 mm, where the change is about one grey level. In 6 of the 17 misses the dent hypothesis converged on the true location, but its evidence stayed below $\tau = 1.52$. The threshold is held up by appearance changes: stains account for 8 of the 9 false alarms, with stain-only fit gains up to 4.7 (5.6 for one healthy blade under camera error). Lowering τ to 0.54 would reach 70% recall at a 20% false-alarm rate. The misses are therefore the price of the false-alarm guarantee: an inspector receives few spurious damage claims, and dents below about 0.1 mm are reported as unsupported rather than guessed.

Transfer to unseen blade designs. Table 6 applies every method, unchanged, to the six CAD6 blades (456 cases). Separation falls from 0.82 on the Rotor37 sealed test to 0.74 (transfer gap 0.08), with 94% of visible dents found, 67% false alarms and 33% abstention on hidden dents. The threshold was calibrated on Rotor37 only, so the false-alarm rate here measures how far the guarantee survives a change of geometry, not a property the method was fit to. Per blade (Table 7), dents are still ranked above negatives (best AUROC on blade 1 (24 mm, AUROC 0.98), worst on blade 3

Method	AUROC $\uparrow$	Found % $\uparrow$	Located % $\uparrow$	False alarm % $\downarrow$	Abstain (hidden) % $\uparrow$	Depth err. (mm) $\downarrow$	Centre err. (mm) $\downarrow$
Ours (v1, frozen)	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]	0.03	0.13
without the evidence test	0.82 [0.71, 0.90]	84 [68, 95]	70 [56, 84]	32 [21, 43]	45 [36, 54]	0.03	0.68
2D detector, lifted	0.71 [0.63, 0.80]	24 [12, 40]	16 [5, 30]	4 [1, 9]	96 [93, 99]	—	1.82
MLP deformation field [†]	0.74 [0.65, 0.81]	16 [6, 29]	11 [3, 22]	6 [1, 10]	100 [100, 100]	—	1.58
Free-form field [†]	0.63 [0.54, 0.70]	14 [4, 24]	11 [3, 20]	4 [1, 9]	99 [97, 100]	—	4.69
Discrete search	0.53 [0.43, 0.65]	5 [0, 13]	5 [0, 13]	6 [1, 10]	98 [95, 100]	—	1.60
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]	—	—

Table 2. Sealed test (240 cases, 30 unseen Rotor37 blades). Found / located: visible dents ($\geq 50\%$ visible in the two input views) reported / reported within 0.1 span of the true centre. Each method uses its own threshold, calibrated on the calibration split (split-conformal, $\alpha = 0.05$); "without the evidence test" reports every fit. Errors are medians over reported dents. Mean [95% interval, resampling blades]. [†]Reimplementations of the core idea of CADSplat [17], not the official model.

Views	Observable dents	AUROC $\uparrow$	Located % $\uparrow$	Median rel. local err. $\downarrow$
2 views	10/60	0.847	60	0.43
4 views	17/60	0.777	41	1.00
7 views	29/60	0.857	62	0.47

Table 3. More views make more dents observable and the evidence clearer.

Variant	AUROC $\uparrow$	Located % $\uparrow$
Full method	0.855	72
No camera (pose) correction	0.679	72
No stain (colour) model	0.753	83
Whole image instead of interior evidence	0.685	61
No score-test proposals	0.420	6
No low-frequency brightness field	0.846	72

Table 4. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

True depth (mm)	2 views	2 views (same cases)	4 views	7 views
< 0.1	1/10 (1 loc.)	0/4 (0 loc.)	1/5 (0 loc.)	3/9 (3 loc.)
0.1–0.2	7/14 (7 loc.)	1/2 (1 loc.)	2/6 (2 loc.)	7/11 (7 loc.)
> 0.2	12/13 (12 loc.)	4/4 (4 loc.)	4/6 (4 loc.)	8/9 (6 loc.)
All visible dents	20/37 (20 loc.)	5/10 (5 loc.)	7/17 (6 loc.)	18/29 (16 loc.)

Table 5. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
Ours, Rotor37 sealed test	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]
Ours	0.74 [0.67, 0.84]	94 [89, 98]	70 [60, 81]	67 [36, 97]	33 [3, 67]
2D detector, lifted	0.86 [0.83, 0.89]	42 [35, 49]	36 [30, 42]	6 [3, 8]	100 [100, 100]
MLP deformation field	0.73 [0.68, 0.81]	84 [70, 97]	54 [47, 62]	43 [27, 58]	67 [48, 85]
Free-form field	0.60 [0.56, 0.66]	61 [36, 84]	37 [21, 53]	39 [20, 58]	64 [48, 82]
Discrete search	0.65 [0.58, 0.72]	16 [12, 18]	13 [10, 17]	4 [0, 10]	97 [94, 100]
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]
Ours v1 (frozen), held-out blades 3–6	0.72 [0.65, 0.90]	90 [85, 95]	67 [55, 79]	51 [10, 91]	50 [9, 91]
Ours v1.1, held-out blades 3–6	0.95 [0.93, 0.98]	86 [77, 95]	83 [76, 90]	4 [0, 11]	98 [93, 100]
Ours, CAD6 leave-one-blade-out τ	0.70 [0.63, 0.84]	13 [3, 26]	6 [1, 13]	4 [0, 11]	95 [89, 100]

Table 6. Transfer to six unseen real blade designs (CAD6, 456 cases; dents 0.1–1.0 mm deep). Every method is frozen: shader trained on Rotor37 only, thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

Blade	Airfoil span (mm)	Cases	Visible dents	AUROC	Found (%)	False alarm (%)
1	24	76	31	0.98	100	100
2	26	76	31	0.72	100	100
3	31	76	31	0.64	97	100
4	40	76	31	0.66	94	82
5	52	76	31	0.93	84	6
6	70	76	31	0.86	87	15

Table 7. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

(31 mm, AUROC 0.64)), but the frozen threshold fails on the smaller blades: false alarms fall with airfoil span (Spearman $\rho = -0.88$, $p = 0.02$). Across blades, airfoil span and detection rate have Spearman $\rho = -0.93$ ($p = 0.01$; six blades, so weak evidence): 99% of visible dents were found on the three smaller blades and 88% on the three larger ones. Because dents are specified in millimetres, the same dent is relatively deeper on a small blade; detection tracks depth relative to the airfoil span ($\rho = 0.10$) more strongly than depth in millimetres ($\rho = 0.02$), rising from 92% in the shallowest relative-depth tertile to 97% in the deepest. Diagnosis and fix (v1.1). On a healthy CAD6 capture whose image is identical to the reference, the silhouette pose fit moves a correct camera by about 0.3° and a spurious dent absorbs the misregistration; under real camera error, the image-space reference bias — large on unseen geometry because the shader was trained on Rotor37 only — stays fixed while the blade moves. v1.1 corrects both: it applies the reference correction to the silhouette as well as to shading, and it reprojects the shading correction through the surface so that it moves with the blade (identical to v1 at the nominal pose). Both changes were developed on CAD6 blades 1–2 only and frozen, with a threshold recalibrated on the Rotor37 calibration split, before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%). Recalibrating the threshold on CAD6 negatives, leave-one-blade-out, gives 13% found at 4%

Variant (CAD6, 48 cases)	AUROC	Found (%)	False alarm (%)
Full method	0.71 [0.62, 0.83]	92 [83, 100]	78 [56, 100]
No camera correction	0.90 [0.81, 0.98]	92 [83, 100]	56 [33, 67]
No stain model	0.73 [0.64, 0.84]	96 [88, 100]	89 [78, 100]
Random starts	0.45 [0.37, 0.56]	38 [29, 46]	44 [22, 67]
Fixed grid of starts	0.50 [0.41, 0.62]	33 [17, 54]	44 [11, 78]
Whole image	0.56 [0.44, 0.71]	75 [58, 92]	61 [50, 67]

Table 8. Ablations on CAD6 (frozen settings, fixed 48-case subset).

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
v1 (frozen)	0.79 [0.68, 0.88]	54 [40, 69]	46 [32, 60]	6 [1, 11]	88 [81, 93]
v1.1 (camera-check + correction fix)	0.80 [0.70, 0.88]	52 [38, 67]	46 [32, 61]	8 [2, 14]	88 [81, 93]

Table 9. Second sealed test: 30 further Rotor37 blades (67–96), 240 cases, unused by any earlier experiment. Each method ran once with its frozen threshold.

false alarms: the cost of transferring a threshold across blade designs.

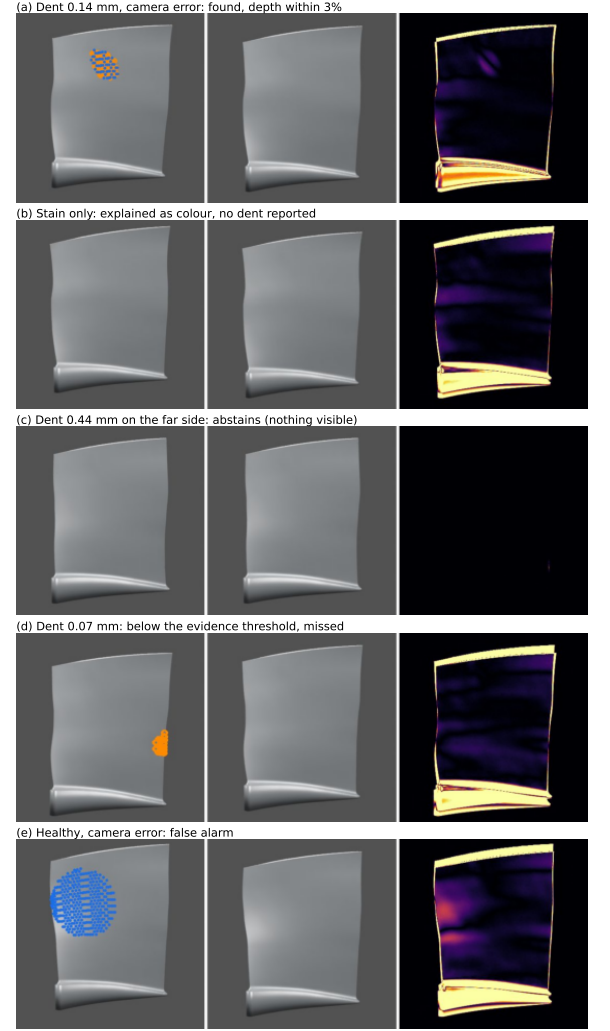
A pre-registered negative result (v2). Targeting the remaining false alarms, we pre-registered three changes and a rule on development blades: a smooth surface appearance field shared by both hypotheses (A), a dent-size prior (D), or both (E), adopted only if recall at matched false-alarm rate improved by at least five points. None did: D matched v1 (67% recall, AUROC 0.87 vs. 0.85), while A and E raised negative fit gains so much that recall fell to 6%; the extra appearance freedom helped the dent hypothesis as much as the null. The change that does reduce false alarms is v1.1’s reference correction of silhouette and shading (above), which also lowers the worst development false-alarm gains on Rotor37 (2.46 to 0.09).

Physical validation (pending real validation).

Pending real validation. A 2× print of CAD6 blade 5 will be dented with steel balls (target depths 0.3, 0.7 and 1.2 mm), photographed with marker-based poses before and after denting, and measured by structured-light scan (or a 0.01 mm dial gauge). This section is reserved for those results; the rest of the paper does not depend on it.

Qualitative results and failures. Figure 2 shows typical outcomes, including a miss and a false alarm.

Real objects. No public dataset pairs real blade photographs with measured damage geometry, so we use Sim3D [7]: six wooden stools scanned with an industrial structured-light sensor before and after damage, with calibrated photographs and expert defect annotations. The healthy scan replaces the CAD model,



Left: input view with true dent (orange rings) and reported dent (blue). Middle: healthy reference. Right: |difference| x10.

Figure 2. Sealed-test outcomes. (a) A 0.14mm dent under camera error is found and measured. (b) A stain is explained as an albedo change. (c) A dent facing away from both cameras yields no evidence and the method abstains. (d) A 0.07 mm dent stays below the threshold. (e) A healthy blade under camera error is reported as dented, the residual failure mode.

the healthy photograph from the same scanner position serves as the reference after 2D alignment, and the measured scan difference is the truth; annotated regions with no measured shape change (paint, contamination) count as negatives. On these textured objects our method does not separate physical defects from undamaged areas (Table 10). The residual between sessions is dominated by wood-grain misregistration and specular changes that the shader, trained on 12 photographs per object, does not model. We report this as a failure case: the evidence test is only as

Objects / defects / negatives	6 / 22 / 31
AUROC (defect vs. undamaged)	0.540
Detected (leave-one-object-out)	0/22
Located within 15 mm	0/22
False alarms	1/31

Table 10. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.

reliable as the appearance model behind it.

6. Limitations and Conclusion

Our positive results use rendered images of public compressor geometry with synthetic damage. On real scanned objects (SiM3D) the method did not separate physical defects from undamaged areas; real-blade validation with measured damage remains future work, and an appearance model that holds on real materials is its main prerequisite. The dent model has one dent per blade, and edge damage such as nicks and material loss is out of scope. The conformal guarantee holds for negatives exchangeable with the calibration set; a new camera rig or lighting setup requires recalibration. The method answers whether these views support a physical change; it does not certify airworthiness or replace an inspector.

We framed blade damage estimation as a decision with a stated level of evidence: report a dent, with its location and depth, only when a dent explains the images significantly better than every non-physical explanation, and abstain otherwise. On rendered Rotor37 blades this cuts the false-claim rate of an unconstrained fit by two thirds (32% to 10%) while keeping most detectable dents, and it outperforms CAD-prior deformation fields and a lifted 2D detector, whose lack of an appearance model forces their thresholds so high that few dents survive. Recall is bounded by what the cameras see: almost all misses are shallower than 0.1–0.2 mm, and seven views instead of two locate nearly three times as many dents at a lower false-alarm rate. Transfer to new blade designs exposed two assumptions that only held for the training family; making the healthy reference follow the silhouette and the camera removed the failure without retraining and without cost on Rotor37. Physical validation on a printed, dented and scanned blade (Section 5) is the next step.

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